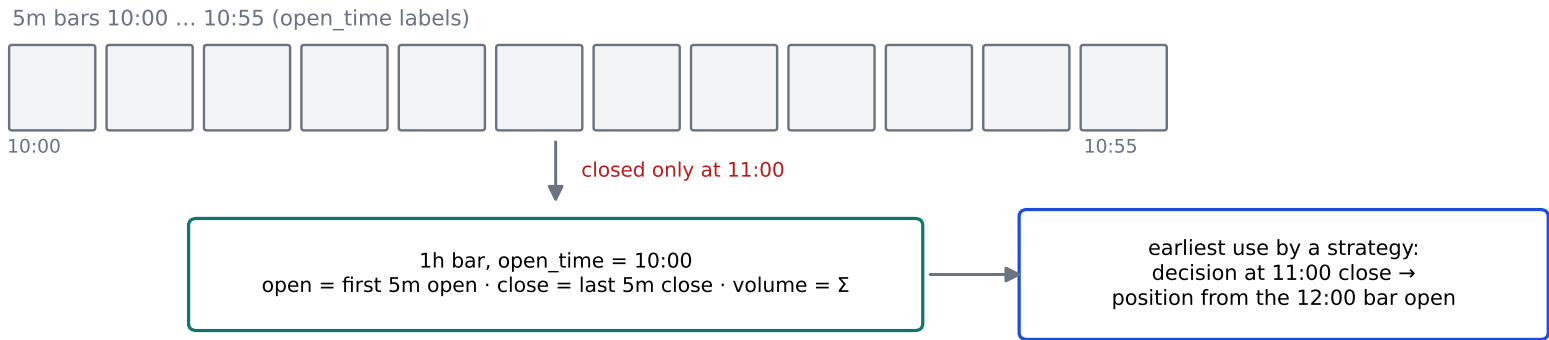
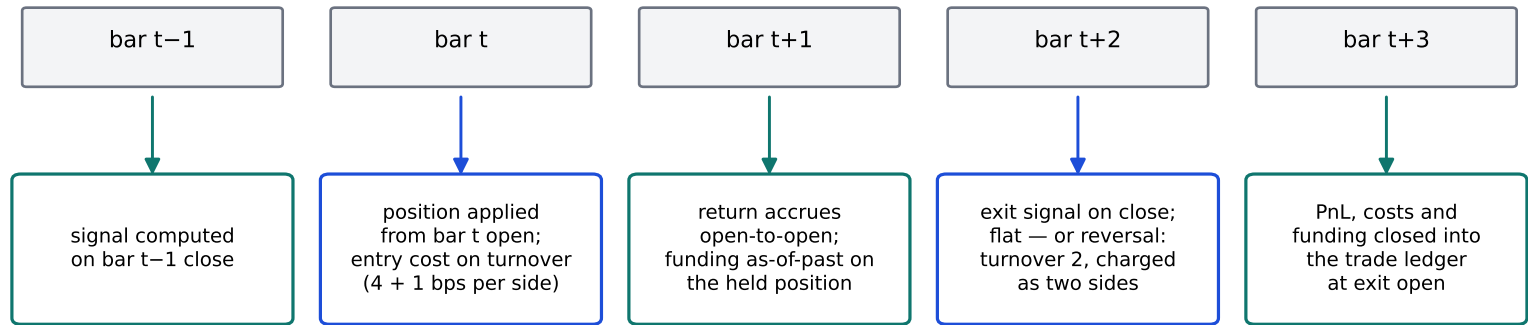


A · Causal aggregation: when a 1h bar becomes observable



B · Next-open execution: one complete trade through the ledger



Invariant: no quantity computed on bar t can influence the position held during bar t. The last bar is dropped (its open-to-open return is unknowable). Pinned-output unit tests fix these numbers exactly.